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The drivers and predictability of wildfire re-burns in the western United States (US)

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E-mail: ksolander@lanl.gov**Keywords:** monitoring trends in burned severity (MTBS), wildland–urban interface (WUI), machine learning (ML), wildfire re-burns

Abstract

Evidence is mounting that the effectiveness of using prescribed burns as a management tactic may be diminishing due to the higher incidence of wildfire re-burns. The development of predictive models of re-burns is thus essential to better understand their primary drivers so that forest management practices can be updated to account for these events. First, we assess the potential for human activity as a driver of re-burns by evaluating re-burn trends both within and outside of the wildland–urban interface (WUI) of the western US. Next, we investigate the predictability of re-burns through the application of both random forest and the explanatory machine learning non-negative matrix factorization using *k*-means clustering (NMFk) algorithms to predict re-burn occurrence over California based on a number of climate factors. Our findings indicate that while most states showed increasing trends within the WUI when trends were conducted over longer moving windows (e.g. 20 years), California was the only state where the rate of increase was consistently higher in the WUI, indicating a stronger potential for human activity as a driver in that location. Furthermore, we find model performance was found to be robust over most of California (Testing F1 scores = 0.688), although results were highly variable based on EPA level III Ecoregion (F1 scores = 0.0–0.778). Insights provided from this study will lead to a better understanding of climate and human activity drivers of re-burns and how these vary at broad spatial scales so that improvements in forest management practices can be tuned according to the level of change that is expected for a given region.

1. Introduction

Wildfire re-burns can be thought of as the area of overlap between burn area perimeters of multiple wildfires that occurred over some time interval. In the western continental United States (US), re-burns have been found to be fairly widespread, with over 7.6% of wildland forests having burned multiple times between 1984 and 2016 (Buma *et al* 2020). Over longer timescales (e.g. 10–25 years), re-burns have been found to be increasing just as rapidly as single-burn wildfires. The increased occurrence of re-burns as an agent of change across western US landscapes suggests that they will continue to be an important part of ecosystem functioning, and more work must be done to improve their understanding so forest management practices can be updated as appropriate.

Re-burns tend to have exacerbated impacts on ecosystem function relative to single burn wildfires due to the inability of vegetation and animal species to adapt to higher-frequency wildfires (Walker *et al* 2018). For example, re-burns have caused an increase in conversion of forests to shrubs or grasslands in both the western US (Buma *et al* 2020) and Patagonia (Paritsis *et al* 2015) regions. The higher fire return intervals brought on by re-burns are expected to substantially change forest age structure and composition (Hart *et al* 2019) as well as lead to failures in re-seeding (Buma *et al* 2013, Bowman *et al* 2014, Stevens-Rumann and Morgan 2019). Given that the warmer and drier climate expected for much of the western US is already

hampering the ability of trees to re-seed, re-burns will likely exacerbate these effects and further changes in forest structure and dynamics (Enright *et al* 2015, Halofsky *et al* 2020). Re-burns have reduced the capacity for tree regeneration, causing a delay in aboveground carbon stores and thus have the potential to result in additional major changes in forest structure and function. Already, the ability of trees to undergo epicormic or basal resprouting has been diminished due to re-burning, leading to greater forest structure vulnerabilities following a wildfire (Fairman *et al* 2019).

Given the substantial number of impacts that re-burns have had on ecosystem structure and function, a number of studies have begun to investigate the primary vegetative and climatic drivers of re-burns. For instance, a random forest model identified mean daytime temperature, relative humidity and shrub cover as being the most important predictors of re-burn severity in forest plots located in Northern California (Coppoletta *et al* 2016). Vegetation was found to have a stronger control on re-burn severity than weather conditions at longer timescales (e.g. >19 years) for sites in the Klamath Mountains Ecoregion of Northern California (Grabinski *et al* 2017). Extreme warm and dry months were found to be consistently associated with higher re-burn severity and maximum temperature was found to explain the greatest variance in re-burn severity for re-burns with shorter return intervals (e.g. 2–14 years). In addition, vegetation characteristics such as the occurrence of snags, logs and dense, low-lying shrubs were determined to provide ideal conditions for high re-burn severity.

Across larger spatial scales, different climatic and vegetative controls on fire return intervals likely play a role in the spatial variability of the occurrence of re-burns. For example, the highest concentration of re-burns was found to be located in Southern California and the southwestern desert mountains, as well as the northern Rocky Mountains of Idaho (Buma *et al* 2020). However, the higher occurrence of re-burns in these regions may not be driven by the same features. The Idaho portion of the Rockies is characterized by the largest increase in area burned, where large burn years are typically preceded by months of drought; whereas large burn years in the hotter and drier southwest are often preceded by months of elevated moisture that promote the build-up of fine fuels (Little *et al* 2009). These spatial differences suggest that while re-burns in dry climates may be precipitated by a period of elevated moisture (e.g. fueled by an uptick in fuels), those in moist climates are precipitated by longer periods of drought (e.g. fueled by a drop in moisture). These spatial differences are known to play out in fire return intervals, which are strongly related to re-burns. For example, the fire return interval in the hot and dry Southwest US is estimated at 15 years, while this number more than doubles to 33 years in Montana and boreal forests of Canada where conditions of elevated moisture tend to persist longer to prevent new fire ignitions (Parks *et al* 2018). This finding points to how the immunity of the landscape to wildfires is much greater in forests where moisture limitations tend to be less frequent.

Although the aforementioned studies have investigated a number of possible drivers of re-burns and their spatial variability to quantify their importance for specific plots to the sub-region scale, there has been little research on quantifying the impact of multiple drivers to investigate their importance for re-burns over larger regions. First, to better understand the role of humans in the occurrence of re-burns, we investigate the difference in the magnitude of trends in re-burns throughout the Western US and compare this with the same trends that occur within the wildland–urban interface (WUI). Although re-burn trends have been comprehensively explored in a previous study (e.g. Buma *et al* 2020), this investigation only included re-burns that occurred in wildland forests and thus the potential role of humans in these trends was largely omitted. Further consideration of the human component is necessary given the prevalence of human-caused wildfires in the Western US (Balch *et al* 2017). Next, we combine a gridded dataset of climate and environmental variables with machine learning (ML) techniques to investigate the importance of multiple climate variables as a predictor. Although numerous studies have explored the relationships between climate and wildfire over large areas (Little *et al* 2009, Abatzoglou and Kolden 2013, Williams *et al* 2019) here, we strive to explore these relationships specifically for re-burns. The model is tested over California due to the higher frequency of re-burns observed in this state (Buma *et al* 2020). Model performance is further tested over the US Environmental Protection Agency (EPA) Level III Ecoregions given the diversity of ecosystems and climate in this area that are likely to explain a high degree of differences in model performance.

2. Methods

Re-burns are calculated using the Landsat-based Monitoring Trends in Burn Severity (MTBS) data product, which is maintained and updated by the US Geological Survey Center for Earth Resources Observation and Science and the USDA Forest Service Geospatial Technology and Applications Center. The MTBS was established in 2006 to evaluate the location, extent and severity of wildfires using Landsat-based estimates of wildfire burn areas. Initially, MTBS was based exclusively on the differenced Normalized Burn Ratio (dNBR), but has since been expanded to include the Relativized dNBR, which is an estimate of the relative magnitude of change and removes the bias associated with pre-fire vegetation conditions (Miller and Thode, 2007). As

of October 2019, the MTBS dataset included 22 969 fires. Within the Western US, this dataset consists of only fires above 405 ha and does not include prescribed fires (Picotte *et al* 2020).

Previous studies have indicated that errors due to the incorrect inclusion of pixels into MTBS that were not actually burned were found to be 15%–18% of the total surface area, while errors of exclusion of pixels that were actually burned range anywhere from 0%–45% of the total surface area (Picotte *et al* 2020). These errors tend to result due to differences in the time since wildfire occurred versus the time when the satellite image was generated (Picotte and Robertson 2010), terrain complexity (Kolden and Weisberg 2007) and vegetation composition (Vanderhoof *et al* 2017). In spite of these issues with misclassification, MTBS still represents the longest spatially comprehensive, high resolution, remote sensing-based fire history mapping dataset that is available for the Western US (Buma *et al* 2020). Given these characteristics, MTBS data were deemed to be ideal for training the ML models and conducting the trend analysis that was used in this study.

Re-burn areas were calculated for the 11 states that comprise the continental Western US for each year by finding the intersection of wildfire burn perimeters obtained from MTBS during 1984–2018. The resulting data product displays the total wildfire re-burn perimeters across the Western US. No distinction was made for how many times a given area re-burned (e.g. two times versus more than two times), given that areas that burn more than two times occur at a far lower rate than areas that burn only twice and thus would be difficult to extract meaningful information on climate drivers or trends for areas that burned greater than two times (Buma *et al* 2020). As such, an area that burned multiple times was simply reported as a re-burn area.

The trend analysis was conducted by aggregating all the re-burn perimeter data by area per year for the 11 Western US states. Data were then binned into 5–10- and 20-year moving windows and trends were estimated over the resulting data using the Mann–Kendall trend analysis for detecting monotonic trends (Mann 1945, Kendall 1975). The statistical significance ($p < 0.05$) of the trends for each of the three moving windows within each state was reported. For this calculation, the adjusted p -value based on an effective sample size to account for autocorrelation was used (Hamed and Rao 1998). This sequence of steps was then repeated for only those re-burns that occurred within the WUI of each state, which was assumed to include both intermix and interface WUI. Intermix WUI is represented by those areas with a housing density greater than 6.17 houses per km² and at least 50% of the area is covered by wildland vegetation. Interface WUI represents areas with higher settlement densities and less than 50% wildland vegetation coverage that also lie within at least 2.4 km of heavily forested areas (e.g. areas with greater than 75% wildland vegetation covering more than 5 km²) (Radeloff *et al* 2018).

Climate drivers in California for use in the ML algorithms were derived using gridded, near-surface climate variables that included monthly and annual values of canopy moisture, soil moisture, precipitation, runoff, snow water equivalent (SWE), max temperature, min temperature, wind speed, and total evapotranspiration (Livneh *et al* 2013). These data were derived from 20 000 National Oceanic and Atmospheric Administration Cooperative Observer stations, as well as models and algorithms. The latest version of the data represents an update to an earlier version (Maurer *et al* 2002) that includes increased spatial resolution to 1/16th °, an extended period of analysis and an updated version of the variable infiltration capacity model to estimate hydrologic variables such as SWE. Comparisons of the two data products were found to be largely consistent, with main differences occurring over mountainous areas of the Western US. The data has been applied to a wide range of studies used to assess water and energy budgets, droughts and climate change assessments (Livneh *et al* 2013).

Additionally, we used historical global climate model (GCM) output that was downscaled to 4 km using Multivariate Adaptive Constructed Analogs (MACA) (Abatzoglou and Brown 2012). The GCM used in this simulated dataset consisted of the bcc-csm1-1 m model (Taylor *et al* 2012) from the CMIP5 suite of simulations (Zhou *et al* 2014). This data included annual maximum temperature, minimum temperature, wind speed, and precipitation fields and was used given the potential to use projected climate data for future work. In recognition that there is no temporal correspondence between the GCM simulated data and re-burn occurrence and because we only attempted to predict the entire set of re-burns that occurred over the 1984–2018 observation period, annual means of these data were used to predict the occurrence of re-burns over the 35 year observation period. By including the historical simulated data from the GCM along with the meteorological station derived data that largely consisted of the same variables, our aim is to showcase the difference between the two. Prior to implementation in the non-negative matrix factorization using k -means clustering (NMFk) algorithm, the MACA-derived GCM data was resampled to the larger spatial scale of the Livneh *et al* (2013) dataset for use in this study. A summary of the data parameters, sources, and resolutions can be found in table 1.

Prominent climate drivers contributing to re-burns were determined using the unsupervised ML algorithm NMFk (Alexandrov and Vesselinov 2014). NMFk is a blind source separation technique for the automated extraction of signals present in complex data. NMFk works through decomposing a data matrix X into a feature matrix $W \in \mathbb{R}^{n \times k}$ and a mixing matrix $H \in \mathbb{R}^{k \times m}$ such that $X = W \times H$. In our application m is

Table 1. Summary information for the climate drivers that were used in this study.

Variable	Temporal Range (Frequency)	Native Spatial Resolution	Source
Reburn Weights	1984–2018 (composite)	>405 ha	Monitoring trends in burn severity (MTBS)
Canopy Moisture	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Soil Moisture	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Precipitation	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Runoff	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Snow Water Equivalent	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Temperature (max)	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Temperature (min)	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Total Evapotranspiration	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Wind Speed	1950–2013 (monthly)	1/16	Livneh <i>et al</i> 2013 CONUS near-surface derived meteorology
Precipitation	1950–2020 (annual)	4 km	Abatzoglou and Brown 2012 MACA downscaled global climate model
Temperature (min)	1950–2020 (annual)	4 km	Abatzoglou and Brown 2012 MACA downscaled global climate model
Temperature (max)	1950–2020 (annual)	4 km	Abatzoglou and Brown 2012 MACA downscaled global climate model
Wind Speed	1950–2020 (annual)	4 km	Abatzoglou and Brown 2012 MACA downscaled global climate model

the number of grid-cells and n is the number of climate, hydrologic and wildfire-based input parameters we are using as inputs to NMFk. The matrix deconstruction process is able to extract an identifiable number of signals, k , within the data that can be used to reconstruct the original data when multiplied by the mixing matrix H , which provides weights to each parameter based on their importance to each signal. NMFk solves for different solutions sets for different values of k ranging from 2 to a user-defined value of d . k -means clustering is then used to determine how well a set of extracted features, k , is at accurately describing the original data based on several statistical measures. These measures include robustness, which is the degree to which a model performs as well when using new data versus training data; as well as the cluster silhouette, which measures how similar an object is to its own cluster. The solution clusters are obtained by executing the NMFk analyses with multiple random initial guesses, which was set to 1000 for our purposes. One of the novelties of NMFk is the ability to automatically estimate the optimal number of signals (K^{opt}) present in the data using k -means clustering.

Here, we first apply NMFk to categorize the main climate and hydrologic signals present in California and how they relate to the re-burned areas as determined by the MTBS dataset. We include the MACA-derived climate data (temperature, precipitation, wind), the Livneh *et al* (2013) climate and hydrologic data (evapotranspiration, streamflow, runoff, rainfall, SWE, etc.), and the MTBS data as inputs to the NMFk algorithm. For this analysis, the MTBS re-burn data were resampled to a common 1/16th ° resolution by calculating the percent area re-burn within the larger grid cell over the entire study period (e.g. 1984–2018), creating the parameter ‘mtbs_weights’ discussed below. Additionally, we computed basic statistics (mean, maximum, minimum, standard deviation, quartiles) for each variable, which are then added to the input dataset along with the raw climate data. Each feature of the input data is normalized between 0 and 1 to enable the NMFk algorithm to decompose the input data into constituent signals, which are subsequently assigned weights for each input feature. The MTBS re-burn weights were enhanced by a factor of 7 to strengthen the separation of signals that are influenced by re-burns. Through iteration, we found that a factor of 7 was large enough to separate a distinct re-burn signal, but not so large that other factors were not included in that signal. This allows us to develop a re-burn signal that includes other input parameters that contribute to that signal as well as what climate and hydrologic signals are not seen as contributing to re-burns.

Next, the NMFk algorithm was applied to the climate dataset only in locations where re-burns have occurred according to the MTBS dataset. This allows determination of how climate drivers differ exclusively among re-burn events. Application of the NMFk algorithm to climate data co-located with re-burns proceeded in the same way in which it is described in the previous paragraph when applied to climate data over all of California.

In addition to the NMFk algorithm, a random forest classification model was used to predict re-burn areas across California over the entire study period (1984–2018) using the same climate datasets mentioned previously as inputs. Random forest classification models work by partitioning model predictions into a series of decision trees, the depth and number of which are controlled by model hyperparameters, which are tuned prior to training the algorithm. Model outcomes are arrived at from taking the mean of output from individual trees. These outcomes are used to establish which features are the most important for model prediction. Random forest models are effective at using complex data to model non-linearities and develop robust classification trees. Additionally, support vector machine models were used to classify the re-burned areas with limited success. Thus, we present only the random forest results for simplicity. The US EPA Level III Ecoregions of California were used as additional inputs so the model performance could be evaluated across these subregions. Each region was one-hot encoded as a 1 or 0 for incorporation into the model input dataset. The model was trained using 70% of the California spatial data where the calculated MTBS re-burns (non-weighted) were used as the target variable. Unlike NMFk, which automatically detects the number of features present in the data, random forest imposes no such limits. However, we down-selected to the 200 most important features, which were used to train a new model on the same data to reduce overfitting. Additionally, model hyperparameters (e.g. number of estimators, minimum samples split, minimum sample leafs, maximum depth) were tuned by randomly selecting from an appropriate range of hyperparameters and then iteratively fitting the data using Kfold cross-validation with three folds within the training data. We performed 20 separate fits using different hyperparameter arrangements to determine the hyperparameter values based on the resulting F1-score. The F1-score was reported as the primary performance metric along with the sensitivity and precision scores.

3. Results

The 1984–2018 re-burns for the 11 Western US states are shown in figure 1. The total area covered by re-burns in the Western US was 89 026 km², which represents 3.1% of the total land surface. The highest number of re-burns in terms of total area occurred in Idaho (16 703 km²) and California (16 418 km²), while the lowest number of re-burns in terms of total area occurred in Wyoming (1840 km²). The highest density of re-burns occurred in Idaho, where they comprised 7.7% of the total land surface, while the lowest density occurred in Montana, where they comprised only 0.6% of the total land surface.

Trends in re-burns for the 11 Western US states over the 5-, 10- and 20 year moving windows are shown in figure 2. Results indicate that when the longer 20 year moving windows are considered, trends tend to be stronger within the WUI where 7 of the 11 states had stronger trends indicated by the magnitude of the tau statistic. When the shorter 5 year and 10 year moving windows were considered, only 5 and 2 of the 11 states had stronger trends within the WUI, respectively. Trends were also more likely to be statistically significant ($p < 0.05$) over the longer 20 year moving windows, particularly in the WUI where 7 of the 11 states fell into this category. When shorter moving windows were considered, only 3 (5-year) and 2 (10-year) of the 11 states had statistically significant trends within the WUI. In both of these shorter time periods, there was a greater number of statistically significant trends that occurred when the analysis was conducted over the entire state than within the WUI. Decreasing trends only occurred within the WUI, which was true for 3–5 of the 11 states, depending on which moving window period was considered. However, the decreasing trends were only statistically significant for New Mexico over the 5 year period. California was the only state where trends were consistently stronger within the WUI for all three moving window periods that were considered, although none of these trends were statistically significant. Idaho was the only state that had decreasing trends of re-burns within the WUI for all three moving window periods, despite having the highest surface area and density of re-burns out of the 11 Western states.

The extracted signals of the NMFk algorithm over the entire climate and re-burn dataset are shown over California in figure 3. The top model weights for each signal are shown in table 2. Here, larger parameter weights within a signal indicates that that parameter is important to constructing the signal (or pattern) identified by the NMFk algorithm. Spatially, a larger signal weight for a particular grid-cell indicates that that area is highly associated with the pattern within the input data that is comprised of the input parameters that make up that signal. The optimal number of signals, as determined by the robustness measure of NMFk solutions, is six. Re-burns are represented in the model by the `mtbs_weight` feature. Of the optimal six signals that were identified, signal 2 is most associated with the re-burn perimeters, given that it was the only signal

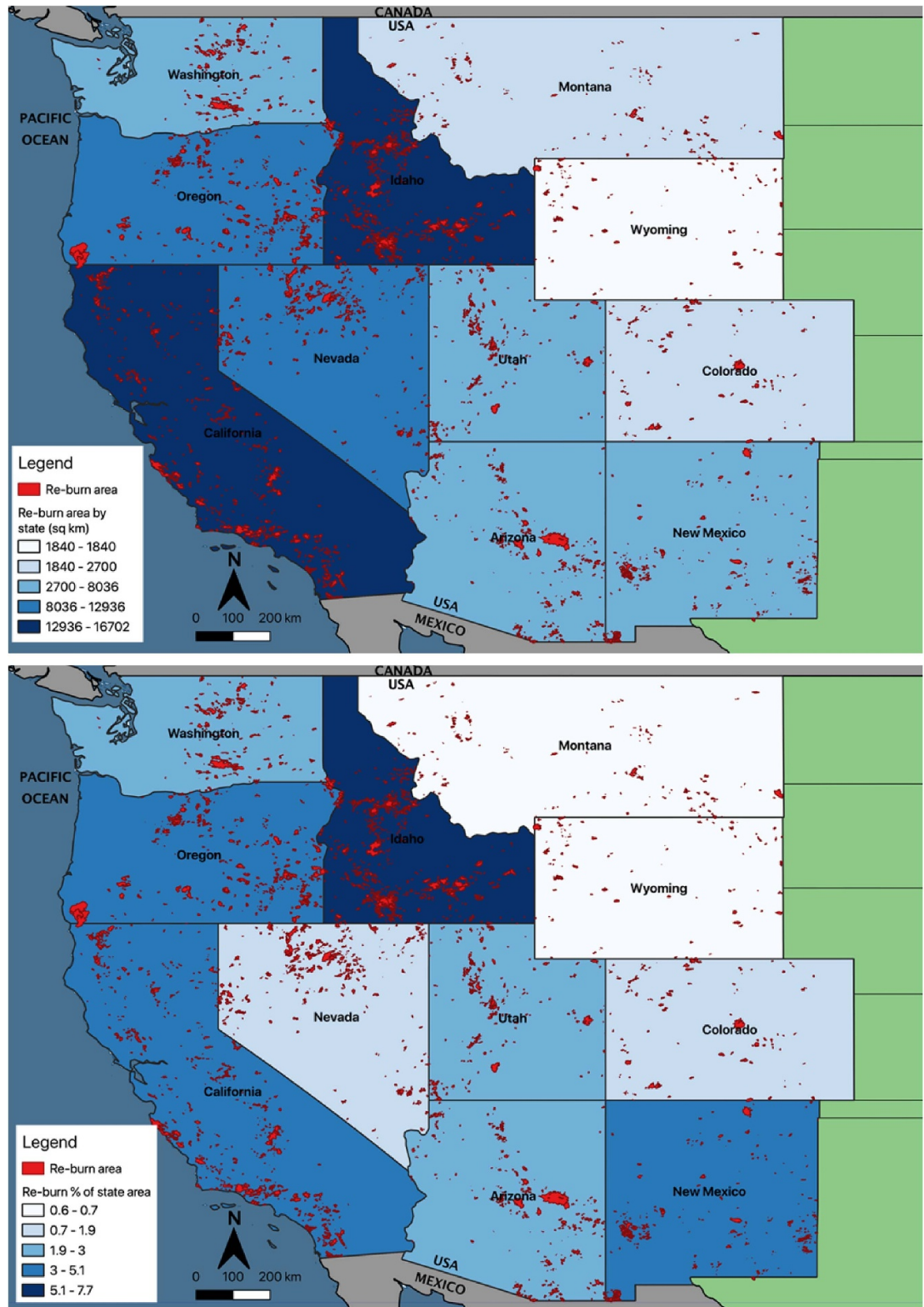


Figure 1. 1984–2018 wildfire re-burn perimeters derived from the MTBS data for in the 11 western US states. States are shaded according to (a) total re-burn area and (b) percent land area coverage that occurred within state boundaries.

where *mtbs_weights* (e.g. re-burns) were included as one of the top features in table 2 and had a normalized weight of 0.80. In addition, the spatial distribution of signal 2 was largely co-located with the MTBS re-burns (figure 3), which was highest along the California coastal mountain and Sierra Nevada foothills regions. Signal 2 is most prominently associated with high evapotranspiration in May–July and moderately associated with high maximum January soil moisture. Re-burns were not determined to be one of the top features for all other signals, as listed in table 2. Signal 1 is associated with both standard deviation in temperature and

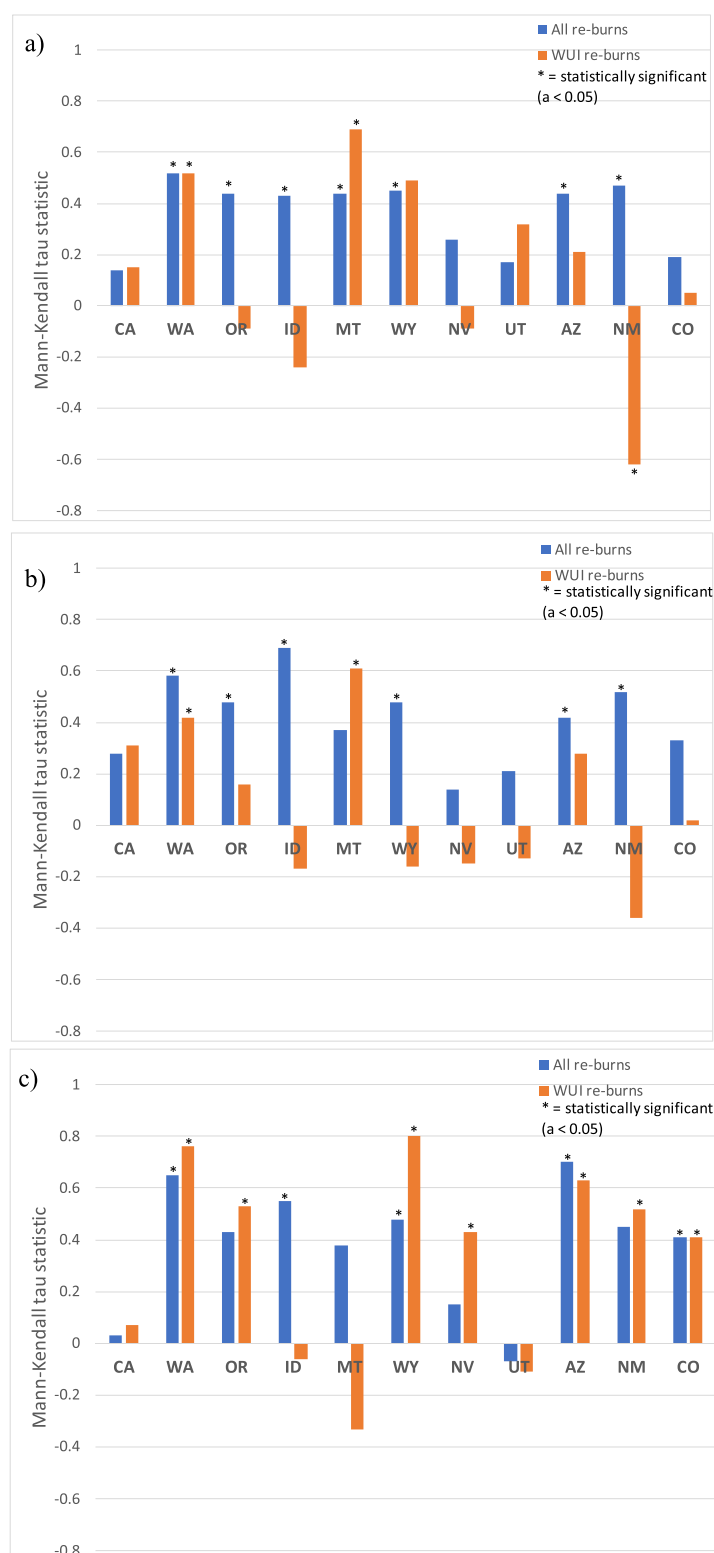
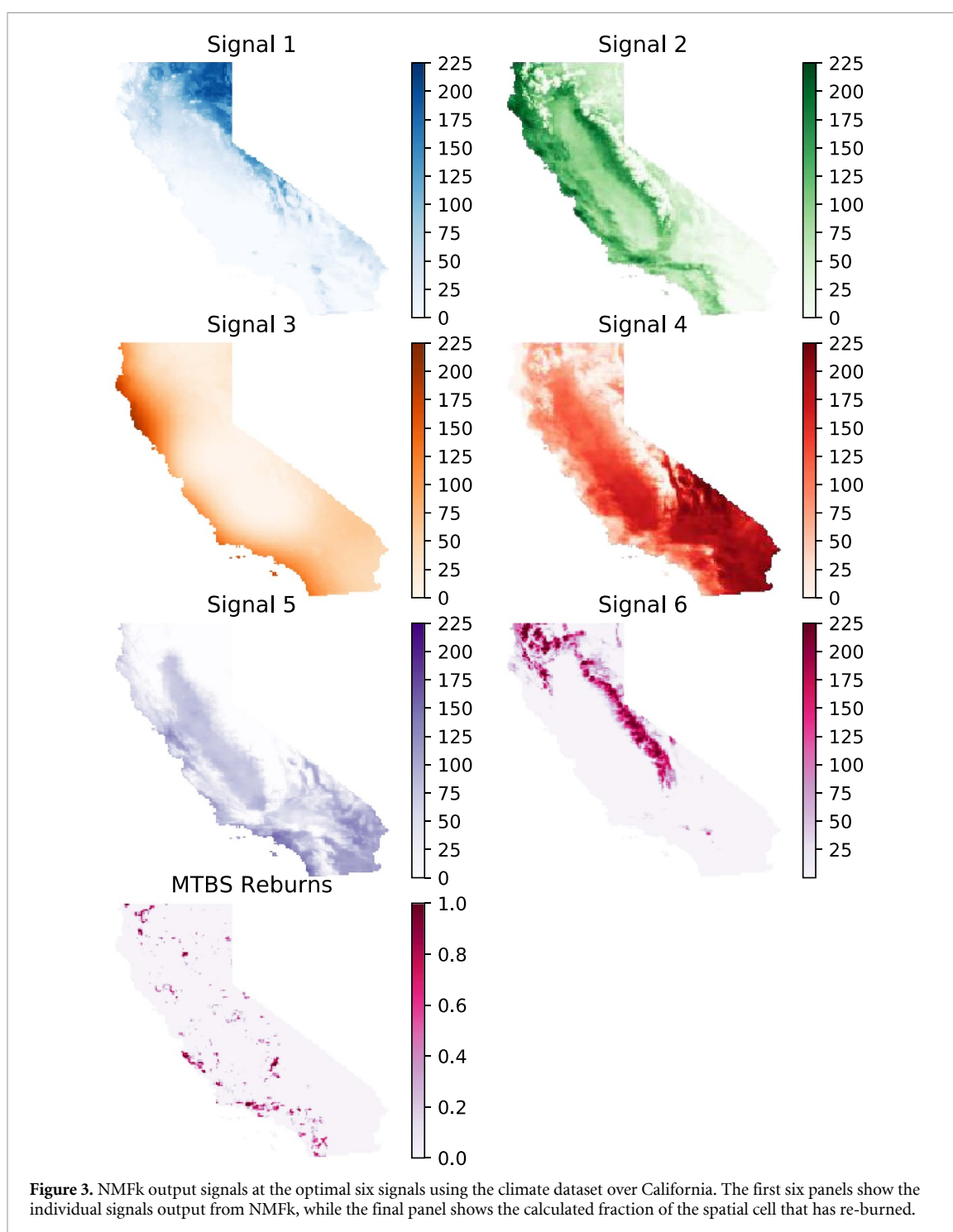


Figure 2. Mann-Kendall tau statistic for trend analysis of re-burns for 11 western US states using (a) 5 year, (b) 10 year and (c) 20 year moving windows. Trends are shown for all re-burns that occurred within each state (blue bars) and only re-burns that occurred within the WUI (orange bars).

summer canopy moisture and is most prevalent in northeastern California. Signal 3 is associated with high wind speed, particularly in September and February–May, and is most prevalent in coastal California. Signal 4 is associated with high maximum temperature and is most prevalent in the California Central Valley and in the more arid desert regions of Southern California. Signal 5 is associated with high minimum temperatures especially during the non-summer seasons. Signal 5 is most prevalent in the Central Valley, Southern



California, and in coastal areas. Finally, signal 6 is associated with high evapotranspiration in July–September and is most prevalent in the high elevation areas of California.

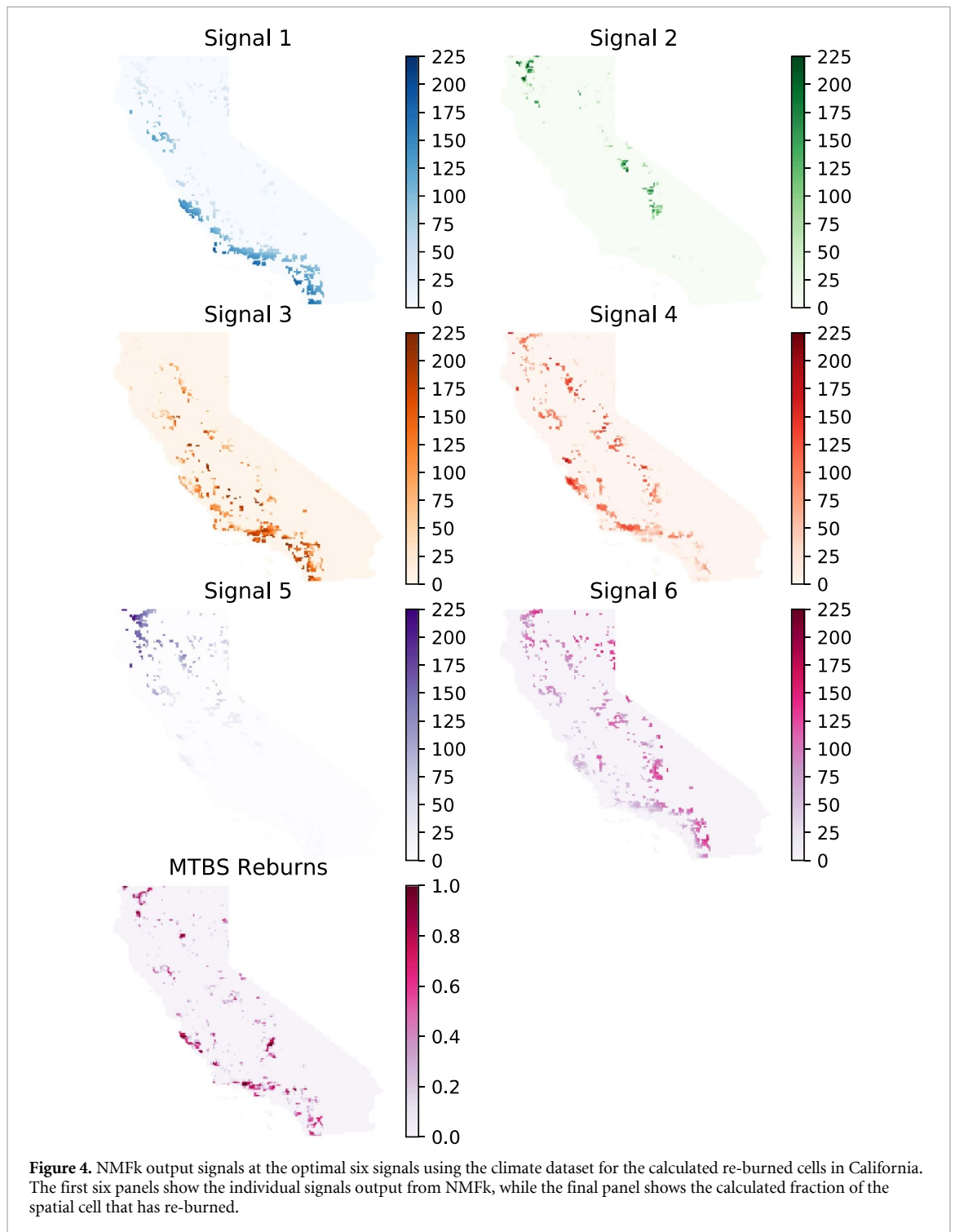
The extracted signals of the NMFk algorithm for only the re-burned areas of California are shown in figure 4. The top model weights for each signal are shown in table 3. The optimal number of signals, as determined by the robustness measure of NMFk solutions, is six. Signal 1 is associated with high wind speed especially during the Spring and is most prevalent in coastal areas of Southern California. Signal 2, which is primarily prevalent in mountainous areas of northwest California and the Sierra Nevada Mountains, is associated with high evapotranspiration in July–September and with the standard deviation of March SWE. Signal 3 is associated with high maximum and standard deviation in winter temperature and is strongest among re-burns located in Southern California and in the foothills to the west of the Central Valley. Signal 4 is associated with high evapotranspiration in April–July and with high standard deviation in January soil moisture. Signal 4 is most prevalent among areas of the central California coast and in some areas of the

Table 2. The top nine features for each of the six NMFk output signals and their corresponding normalized weight for each signal over the entire state of California. The signals correspond to the signals shown in figure 2. Abbreviations are defined in the table footer^{a,b}.

Signal 1			Signal 2			Signal 3			Signal 4			Signal 5			Signal 6		
Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights	Top Features	Weights
Nov temp max std	1.00	May ET mean	1.00	Sep was 1st quartile	1.00	tmax 39	1.00	Feb temp min std	1.00	Jul ET 3rd quartile	1.00	Jul ET 3rd quartile	1.00	Jul ET 3rd quartile	1.00		
Feb temp max std	0.91	May ET 1st quartile	0.94	wind 13	1.00	tmax 23	1.00	Mar temp min 3rd quartile	0.91	Jul ET mean	0.91	Jul ET mean	0.91	Jul ET mean	0.91		
tmin std	0.88	May ET 3rd quartile	0.87	wind 65	0.97	tmax 4	0.97	Mar temp min max	0.86	Aug ET max	0.86	Aug ET max	0.86	Aug ET max	0.86		
Jun canopy_moist std	0.87	Jun ET mean	0.87	Sep was mean	0.96	Jul temp max min	0.96	Oct temp min 1st quartile	0.86	Aug ET 3rd quartile	0.86	Aug ET 3rd quartile	0.86	Aug ET 3rd quartile	0.86		
Sep canopy_moist max	0.86	ET mean	0.85	Mar was mean	0.96	tmax 51	0.96	Jan temp min std	0.85	Jul ET 1st quartile	0.85	Jul ET 1st quartile	0.85	Jul ET 1st quartile	0.85		
May canopy_moist std	0.81	Jul ET max	0.82	Mar was 3rd quartile	0.96	tmax 71	0.96	Apr temp min 1st quartile	0.85	Jul ET max	0.85	Jul ET max	0.85	Jul ET max	0.85		
Jun canopy_moist 3rd quartile	0.76	mtbs_weights	0.80	May was 1st quartile	0.95	tmax 31	0.95	Nov temp min max	0.85	Aug ET mean	0.85	Aug ET mean	0.85	Aug ET mean	0.85		
Aug canopy_moist std	0.75	Jun ET 3rd quartile	0.78	Apr was 1st quartile	0.95	tmax 8	0.95	Oct temp min	0.85	Sep ET max	0.85	Sep ET max	0.85	Sep ET max	0.85		
Jun canopy_moist mean	0.70	Jan soil_moist max	0.77	Feb was min	0.95	tmax 65	0.95	Jan temp max min	0.83	Sep ET 3rd quartile	0.83	Sep ET 3rd quartile	0.83	Sep ET 3rd quartile	0.83		

^a canopy_moist = Canopy Moisture; ET = Evapotranspiration; max = Maximum; min = Minimum; mtbs = Monitoring Trends in Burn Severity; soil_moist = Soil Moisture; std = Standard Deviation; tmax = Maximum Temperature; tmin = Minimum Temperature; was = Wind Speed.

^b tmax 51 = Maximum Temperature for the 51st year (e.g. 2000) of the time series from 1950 to 2020.



foothills of the Cascades in Northern California. Signal 5 is strongest only in mountainous regions of Northwest California and is associated with high standard deviations in winter wind speed and with generally high annual wind speed values. Signal 6 is associated with high Summer SWE and runoff and is prevalent among re-burns in the higher elevation mountainous regions throughout California.

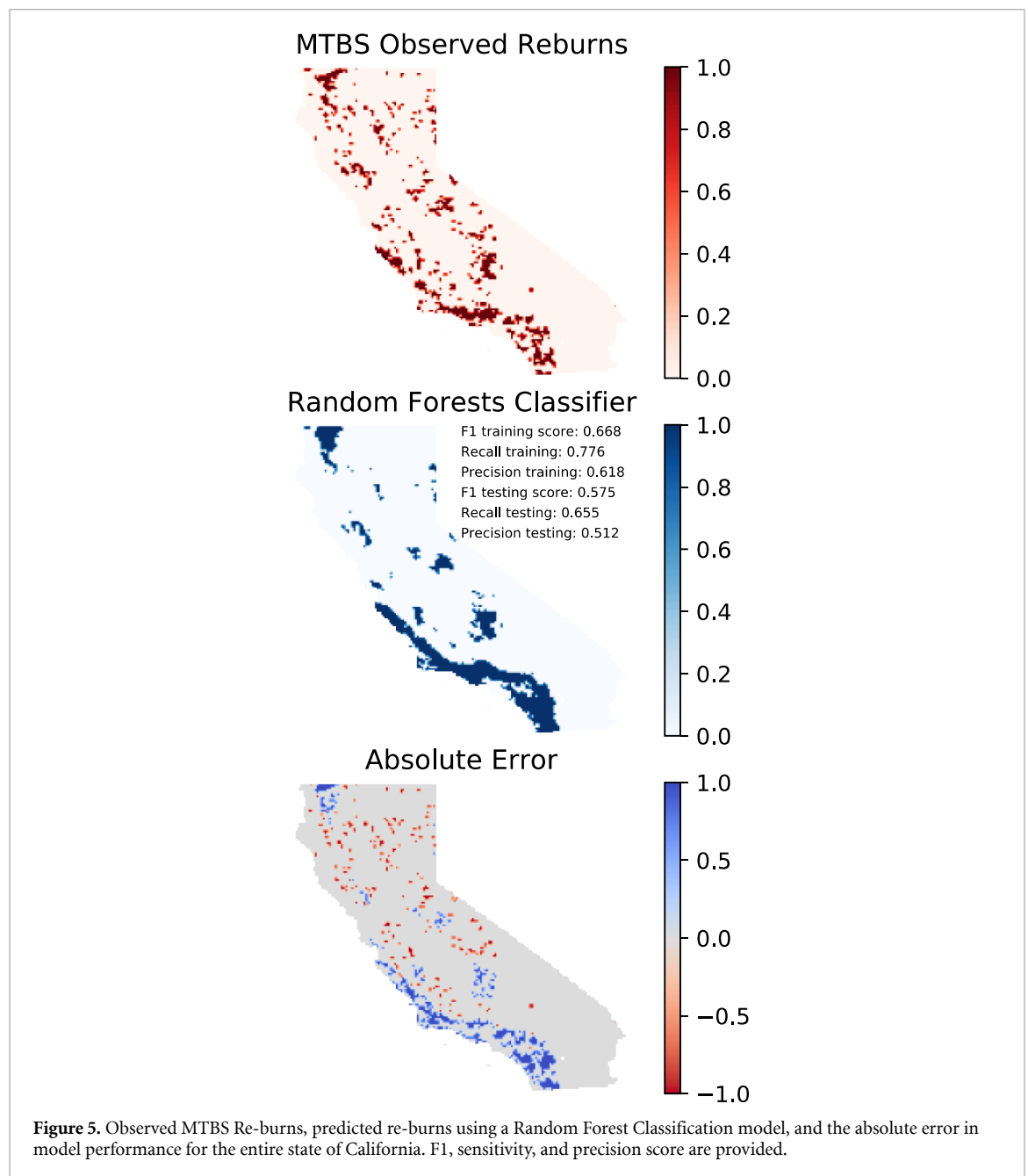
The results of the RF classifier model are shown in figure 5, which shows the predicted and observed re-burns for the state of California, as well as the standard error for these predictions. The final RF model hyperparameters were determined to be 1000 estimators, a max depth of 7, a minimum samples split of 5, and a minimum samples leaf of 2. The model was able to achieve a testing F1 score of 0.596, although some overfitting did occur (training F1 score = 0.695). It is evident that the model over-predicted the size of re-burn perimeters, particularly along the southern and central coasts of California. However, we would not expect the model to capture the exact outlines of re-burn areas as the true perimeter could be related to

Table 3. The top nine features for each of the six NMFk output signals and their corresponding weight for each signal using only the re-burned cells. The signals in correspond to the signals shown in figure 3. Abbreviations are defined in the table footer^{a, b}.

Signal 1			Signal 2			Signal 3			Signal 4			Signal 5			Signal 6		
Top Features	Weights		Top Features	Weights		Top Features	Weights		Top Features	Weights		Top Features	Weights		Top Features	Weights	
wind 7	1.00		Jul ET 3rd quartile	1.00		Jan temp min std	1.00		Jun ET 3rd quartile	1.00		Nov was std	1.00		Oct SWE min	1.000	
May was 1st quartile	0.96		Jul ET mean	0.90		Feb temp min std	0.92		May ET 1st quartile	0.94		wind std	0.98		Sep runoff min	1.000	
wind 64	0.96		Jul ET max	0.83		Dec temp min std	0.91		May ET mean	0.88		wind 4	0.97		Jun SWE min	1.000	
Apr was 1st quartile	0.94		Aug ET max	0.81		Feb temp min max	0.87		May ET 3rd quartile	0.88		Mar was std	0.96		Aug SWE 1st quartile	1.000	
Apr was mean	0.94		Mar SWE std	0.81		Nov temp min max	0.85		Apr ET 1st quartile	0.84		Feb was std	0.94		Jul runoff min	1.000	
Apr was 3rd quartile	0.94		Aug ET 3rd quartile	0.80		Nov temp min 3rd quartile	0.79		Jul ET max	0.79		wind 37	0.85		Sep SWE 1st quartile	1.000	
wind 2	0.94		Jul ET 1st quartile	0.79		Mar temp min 3rd quartile	0.78		Jun ET mean	0.77		Dec was std	0.82		Jul SWE min	1.000	
Mar was 1st quartile	0.93		Jun ET 3rd quartile	0.77		Feb temp min 3rd quartile	0.77		Apr ET min	0.75		wind 29	0.82		temp min	0.997	
wind 15	0.93		Sep ET max	0.73		Dec temp min max	0.77		Jan soil_moist std	0.74		wind 68	0.82		Aug SWE min	0.997	

^a canopy_moist = Canopy Moisture; ET = Evapotranspiration; max = Maximum; min = Minimum; mtbs = Monitoring Trends in Burn Severity; std = Standard Deviation; SWE = Snow Water Equivalent; tmax = Maximum Temperature; tmin = Minimum Temperature; was = Wind Speed.

^b wind 64 = Wind Speed for the 64th year (e.g. 2013) of the time series from 1950 to 2020.



factors extraneous to our input data such as vegetation cover, fire management, and the presence of natural or constructed fire-breaks.

The results of the RF classifier model are broken down by US EPA Level III Ecoregions in table 4. The model performs poorly in some areas where re-burns are sparse, which is denoted by lower precision scores (e.g. < 0.6) and are indicative of a high number of false positives relative to the total number of re-burns. These regions include the Coast Range and Cascades (no predicted re-burns), which had the third and fourth fewest re-burn pixels of the regions represented. However, the model does perform best in other areas where re-burns are less frequent, such as the Sonoran and Central Basin and Range Ecoregions, which had the fewest number of re-burn pixels of the regions represented. The model performs well (accuracy 0.87–0.92) in the Klamath Mountains, Central California Foothills, and Sierra Nevada, where re-burns are well-represented (precision score > 0.6). In the southern California Mountains and Southern California/Baja Coast where we see the highest percentage of reburns (34%–43%), the precision score is low indicating many false positives and the recall score is high indicating a small number of false negatives. The maps provided in figure 5 indicate that this is likely because the model is over-predicting re-burns in these areas. Again, the model has the skill to know that re-burns are occurring in this area, but the climate data alone does not appear to be sufficient to accurately represent the extent of the re-burn areas.

Table 4. Results statistics for the random forest classifier across the entire domain and for each US EPA Level III Ecoregion. Also included are the number of cells, number of burned cells and percent area burned within each ecoregion.

	Total	Coast Range	Central Basin and Range	Mojave Basin and Range	Cascades	Sierra Nevada	Central California Foothills and Coastal Mountains	Central California Valley	Klamath Mountains/California High North Coast Range	Southern California Mountains	Northern Basin and Range	Southern California/ Northern Baja coast	Eastern Cascades Slopes and Foothills
F1 Score	0.660	0.308	0.703	0.615	0	0.671	0.673	0.435	0.653	0.675	0.778	0.694	0.261
Accuracy	0.916	0.977	0.967	0.992	0.945	0.92	0.873	0.979	0.883	0.584	0.963	0.698	0.965
Precision	0.592	0.286	0.929	0.571	0	0.729	0.619	0.833	0.616	0.513	0.875	0.535	1
Recall	0.747	0.333	0.565	0.667	0	0.621	0.736	0.294	0.695	0.983	0.7	0.988	0.15
N Burns	6	6	23	18	21	182	356	34	141	175	10	171	20
N Cells	10 688	385	338	1905	383	1386	2011	1224	889	399	107	494	480
% Burn	0.06	1.56	6.80	0.94	5.48	13.13	17.70	2.78	15.86	43.86	9.35	34.62	4.17

4. Discussion

Our findings related to the spatial variability of re-burn occurrence and trends are supported by several recent studies. For instance, the higher re-burn occurrence and/or densities observed within California and Idaho corroborate with the US EPA Level III Ecoregions that were identified as having higher re-burn percentages (Buma *et al* 2020). Of the eight Level III Ecoregions with elevated re-burn percentages (e.g. > 7%), five were at least partially located in California or Idaho (Sierra Nevada = 7.7%, Central California Foothills and Coastal Mountains = 13.2%, Idaho Batholith = 8.1%, Klamath Mountains = 11.4%, and Sonoran Basin and Range = 16.7%). The consistently stronger increasing trends of re-burns that were found in the WUI only within California are perhaps related to the higher occurrence of human-caused wildfires within that state (85%). Contrastingly, in Idaho where the density of re-burns was still high, only decreasing trends were observed within the WUI possibly due to the lower incidence of human-caused wildfires (31%). This is supported by the discovery that the strongest increasing trends in lightning caused wildfires to have been detected in the North American Desert Ecoregion, which comprises a large portion of Idaho (Balch *et al* 2017). In addition, the decreasing trends that occurred within the WUI in Idaho and elsewhere are perhaps related to the stronger fire suppression efforts that are more likely to take place there to protect critical infrastructure than in more remote locations. Fire suppression expenditures are reported to have increased by 250% from 2000 to 2013 relative to 1985–1999 and a main reason for this increase was the expansion of residential development within the WUI (Clark *et al* 2016). The finding of more robust trends overall, particularly within the WUI, when the longer 20 year moving window is considered is supported by the lack of fuels following a burn that will decrease the likelihood of re-burns over shorter intervals (Buma *et al* 2020). The protection against wildfire in the years following a burn due to the fuels limitation is reported to be particularly notable in the first ten years after a wildfire (Hart *et al* 2019).

Differences in regional climate and associated impacts on the fuels limitation have been identified as an important driver in the frequency of re-burns. For example, in the warm and dry climate of the southwestern US where the likelihood of wildfire re-burns is lower due to fuel limitations, the occurrence of re-burns has been strongly linked to periods of elevated moisture that can boost fuel loads. Forests within these climate zones have been observed to have a short fire return interval of 9–15 years (Parks *et al* 2016, 2018), perhaps because once the vegetation has regenerated following a wildfire, the persistence of elevated moisture over some time period can increase fuel loads above some tipping point that is necessary to carry fire. However, in cooler and more moist climates like the Northwestern US where the likelihood of wildfire re-burns lower due to moisture limitations, decreases in moisture (e.g. droughts) will increase the probability of re-burns (Buma *et al* 2020). Forests within these climates have been observed to have longer fire return intervals of 20–33 years (Parks *et al* 2016, 2018) indicating that the moisture limitations to re-burns in these climate zones are generally stronger than the fuel limitations to re-burns in hot and dry climate zones.

Using NMFk for the analysis over all of California, we were able to derive which climate data is associated with a re-burn signal across all of the re-burned areas in California. ET was found to be a major driver of re-burns across all of California, especially in the late-Spring to early-Summer months. This was evident in signal 2 (table 2), which was the only signal with a strong relationship to the occurrence of re-burns. Next, using only the re-burned areas of the state as inputs to the NMFk algorithm, we were able to disaggregate the re-burned areas to determine how climatic drivers differ spatially within re-burn areas of California. The NMFk results from only the areas of re-burn show that elevated ET in both the Summer and late Spring months comprise the bulk of the features in signals 2 and 4 (table 3), respectively. The strong relationship between re-burns and Spring and Summer ET highlights the importance of antecedent moisture conditions on re-burns given that wildfires typically occur later in the year in California (Swain 2021). We also see strong signals associated with re-burn behavior based on raw wind speed, wind speed variability and seasonal wind speed. This is shown in signal 5 of figure 3, which is associated with re-burns located in Northern California as well as signal 1 of figure 3, which is associated with re-burns located in Southern California. The strong signals associated with wind from these regions may be due to the Diablo and Santa Ana wind events, which are both strongly associated with wildfires (Smith *et al* 2018). Both signal 1 and signal 5 share many of the same wind features. Signal 5, however, is more concerned with high variability in wind speed, while the top contributor of signal 1 is the raw wind speed indicating both are important features for re-burns. The wind speed standard deviation may be linked to the onset of these events, given that their presence is likely to increase the variability of wind speed in the season they occur. NMFk clearly shows that Fall and Winter minimum temperatures are also heavily associated with re-burns in signal 3, which covers re-burns in both northern and Southern California, indicating that the high minimum temperatures brought on by the winds are also important. Snow and hydrology are also important for re-burns in high elevation areas, as evidenced by signal 6. These variables tended to be associated with the Summer and Fall months, which coincide with the peak fire season and underscores the importance of fuel moisture conditions.

The results of the RF classification model show some skill in predicting the general regions where re-burns are most prevalent for the entire study period from 1984 to 2018. However, the model overpredicts re-burn perimeters particularly in Southern California likely due to the lack of representation of key features related to vegetation and fire management that are important drivers of re-burns. Even still, because the model is proficient at identifying the general regions most impacted by re-burns (e.g. Southern and Central California Coast, Sierra Nevada Foothills and Northwest Mountains), the skill at properly identifying the general areas where re-burns occur based on climate data alone shows promise for future applications within a model. This could include understanding how re-burn areas might expand or contract based on future climate change, which could have implications for changes in forest management practices to receive more regular fuel treatments.

As alluded to in the previous paragraph, several potentially important features were omitted from our ML model approaches at predicting re-burns in order to focus our predictors on climate. For example, vegetation characteristics such as composition and structure were shown to be important controls on re-burns at the local or Ecoregion scales (Coppoletta *et al* 2016, Grabinski *et al* 2017). Moreover, the productivity of fine fuels was found to be important at larger, more regional scales (Buma *et al* 2020). Another important feature missing from the current study is the ability to test how human activity may drive re-burn occurrence within the existing ML model framework. Although this is tested through the trend analysis, doing so within our ML model approach would require estimates of WUI through time at a temporal resolution that was comparable to the climate drivers. To the best of our knowledge, this data was not available at the time of writing this manuscript. However, this information could be derived directly through analysis of optical imagery (e.g. Landsat or the Moderate Resolution Imaging Spectroradiometer) and/or approximated by evaluating changes in population growth from census data within the WUI. Both approaches were beyond the scope of this study, but could lay the foundation for future research on this topic.

5. Conclusion

The trend analysis of re-burn occurrence demonstrated that the strength of trends is generally higher within the WUI than outside this region particularly when trends were considered over longer time intervals (e.g. 20 years) due to the fuel limitation for shorter fire return intervals (e.g. <10 years). California was the only state where fuel limitations were consistently stronger within the WUI over all time periods, possibly due to the much higher incidence of human-caused wildfires relative to other states. The applied unsupervised ML algorithm (NMFk) was able to highlight several important climatic features that are most strongly related to the occurrence of re-burns across a large area such as California. These include temperature, particularly during the Fall and Winter months, ET during the Spring and Summer months, raw and standard deviation of wind speed, as well as runoff and SWE particularly during the peak fire season. The random forest classification model offered some skill at predicting general locations of re-burns based on climate data alone, although struggled with some false positives likely due to the lack of some important features related to vegetation or topography that were not incorporated into the model.

The results from this study have implications for the prioritization of fuel treatments such as thinning or prescribed burns by forest managers (Halofsky *et al* 2020). Knowledge on the immunity of a landscape to fire can help managers determine the frequency of retreatments following fuel treatments. This is particularly important for areas treated with prescribed burns, which offer mixed results in effectiveness where mitigation in burn severity can range anywhere from 2- to 30- years post treatment (Prichard *et al* 2017). Moreover, burn severities in re-burns have been found to not necessarily be lower, but generally reflective of the initial fire. These findings coupled with the results of this study suggest that prioritization of more regular treatment within the WUI of California than other states could be more effective in avoiding losses in structures or lives given the higher incidence of re-burns overall (figure 1) and consistently stronger increasing trends in re-burns found in these locations than elsewhere in the state (figure 2). Specific areas within the WUI to target could include those areas where seasonal wind speed and evapotranspiration rates were shown to be strongly related to the occurrence of re-burns (figures 3, 4 and table 4), as well as those areas that were shown to have generally higher re-burn densities according to the observations (figure 1) and model (figure 4). Future work could involve identifying areas of higher risk using this data, running the model using projected climate data to understand changes, the use of vegetation and topographic data to improve existing models, and testing separate ML models over the individual ecoregions to refine model predictions at a smaller scale that could be used as a supplement to the larger model.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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